

ANALYSIS OF AADHAAR ENROLLMENT & UPDATE TRENDS USING UIDAI DATA

A Data-Driven Research Report



Unique Identification Authority of India

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Submitted by

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Executive Summary

This report presents a comprehensive data-driven analysis of Aadhaar enrollment and update trends across India, leveraging official API datasets provided by the Unique Identification Authority of India (UIDAI). The study examines three primary data dimensions – enrollment, biometric updates, and demographic distribution – across states, districts, and age groups.

Key findings reveal significant regional disparities in Aadhaar penetration, with states such as Meghalaya and Delhi leading enrollment activity while many rural and north-eastern states lag behind. Biometric update compliance remains inconsistent across age groups, and demographic analysis exposes systemic service delivery gaps. Actionable policy recommendations are provided to bridge these gaps.

1. Problem Statement

The objective of this research is to identify meaningful patterns, trends, anomalies, and predictive indicators within Aadhaar enrollment and update data provided by UIDAI — and to translate these insights into clear, actionable frameworks that support informed decision-making and systemic improvements in Aadhaar service delivery across India.

2. Approach & Methodology

2.1 Research Approach

Python was employed as the primary analytical tool to uncover meaningful insights from Aadhaar enrollment and update datasets. The analysis focused on understanding footfall patterns at specific enrollment centres, the distribution of services by age group, and state- and district-level disparities in participation.

2.2 Data Cleaning

The raw dataset contained several inconsistencies that required systematic remediation:

- Removal of spelling errors in state and district names
- Correction of districts incorrectly mapped to wrong states
- Standardisation of uniform naming conventions across the dataset
- Handling of missing Aadhaar centre names
- Identification and removal of missing values encoded as the placeholder value "100000" (particularly in pin code fields)
- Consolidation of three separate CSV files into a single dataset of approximately one million records

2.3 Preprocessing

Following cleaning, advanced preprocessing was performed to enable standardised and comparable analysis:

- A derived feature – total enrollment – was created by aggregating relevant enrollment-related columns.
- To address unequal temporal coverage across states, normalisation was applied by computing average total enrollment per state, ensuring temporal consistency and cross-state comparability.

3. Datasets Used

All data used in this analysis was sourced exclusively from official UIDAI API datasets. No external or third-party data sources were employed.

Dataset	Key Field	Age Groups Covered
Aadhaar Enrollment	Date, State, District, Pincode	0-5 yrs, 5-17 yrs, 18+ yrs
Aadhaar Demographic	Date, State, District, Pincode	0-5 yrs, 5-17 yrs
Aadhaar Biometric	Date, State, District, Pincode	0-5 yrs, 5-17 yrs, 18+ yrs

Sample: Enrollment Dataset

date	state	district	pincode	age_0_5	age_5_17	age_18_greater	total_enrollement
2/3/2025	Meghalaya	East Khasi Hills	793121	11	61	37	109
9/3/2025	Karnataka	Bengaluru Urban	560043	14	33	39	86
9/3/2025	Uttar Pradesh	Kanpur Urban	208001	29	82	12	123
9/3/2025	Uttar Pradesh	Aligarh	202133	62	29	15	106
9/3/2025	Karnataka	Bengaluru Urban	560016	14	16	21	51
9/3/2025	Bihar	Sitamarhi	843331	20	49	12	81
9/3/2025	Bihar	Sitamarhi	843330	23	24	42	89
9/3/2025	Uttar Pradesh	Bahraich	271865	26	60	14	100
9/3/2025	Uttar Pradesh	Firozabad	283204	28	26	10	64
9/3/2025	Bihar	East Champaran	845418	30	48	10	88
9/3/2025	Uttar Pradesh	Mahrajganj	273164	31	70	13	114
9/3/2025	Bihar	Sitamarhi	843317	35	94	16	145
9/3/2025	Maharashtra	Chhatrapati Sambhajnagar	431001	42	46	12	100

Figure 1 — Sample rows from the Aadhaar enrollment dataset

Sample: Demographic Update Dataset

date	state	district	pincode	demo_age_5_17	demo_age_17_18	total_demo_age
1/3/2025	Uttar Pradesh	Gorakhpur	273213	49	529	578
1/3/2025	Andhra Pradesh	Chittoor	517132	22	375	397
1/3/2025	Gujarat	Rajkot	360006	65	765	830
1/3/2025	Andhra Pradesh	Srikakulam	532484	24	314	338
1/3/2025	Rajasthan	Udaipur	313801	45	785	830
1/3/2025	Rajasthan	Sikar	332028	28	285	313
1/3/2025	Karnataka	Tumakuru	572201	88	332	420
1/3/2025	Uttar Pradesh	Gorakhpur	273211	61	836	897
1/3/2025	Andhra Pradesh	Kurnool	518313	83	986	1069
1/3/2025	West Bengal	Paschim Medinipur	721148	13	281	294
1/3/2025	Uttar Pradesh	Ghaziपुर	233226	134	1591	1725
1/3/2025	Gujarat	Patan	385360	117	719	836

Figure 2 — Sample rows from the demographic update dataset

Sample: Biometric Update Dataset

A	B	C	D	E	F	G
date	state	district	pincode	bio_age_5_17	bio_age_17_	total_biometric
19-09-2025	Goa	North Goa	403502	0	4	4
19-09-2025	Goa	North Goa	403508	1	4	5
19-09-2025	Goa	North Goa	403513	2	0	2
19-09-2025	Goa	North Goa	403527	2	2	4
19-09-2025	Goa	South Goa	403601	7	3	10
19-09-2025	Goa	South Goa	403602	2	3	5
19-09-2025	Goa	South Goa	403703	2	11	13
19-09-2025	Goa	South Goa	403705	6	4	10
19-09-2025	Goa	South Goa	403708	0	2	2
19-09-2025	Goa	South Goa	403709	1	0	1

Figure 3 — Sample rows from the biometric update dataset

4. Key Findings

4.1 Enrollment Analysis

State-Level Average Enrollment

Meghalaya records the highest average daily enrollment among all states, reflecting strong participation and effective access to enrollment services. States such as Delhi, Bihar, Madhya Pradesh, and Uttar Pradesh also perform relatively well, likely supported by better infrastructure and outreach programmes.

Conversely, a significant number of states and union territories show considerably lower average enrollment figures, indicating limited access, lower awareness, or insufficient enrollment activities — underscoring areas where targeted intervention is most needed.

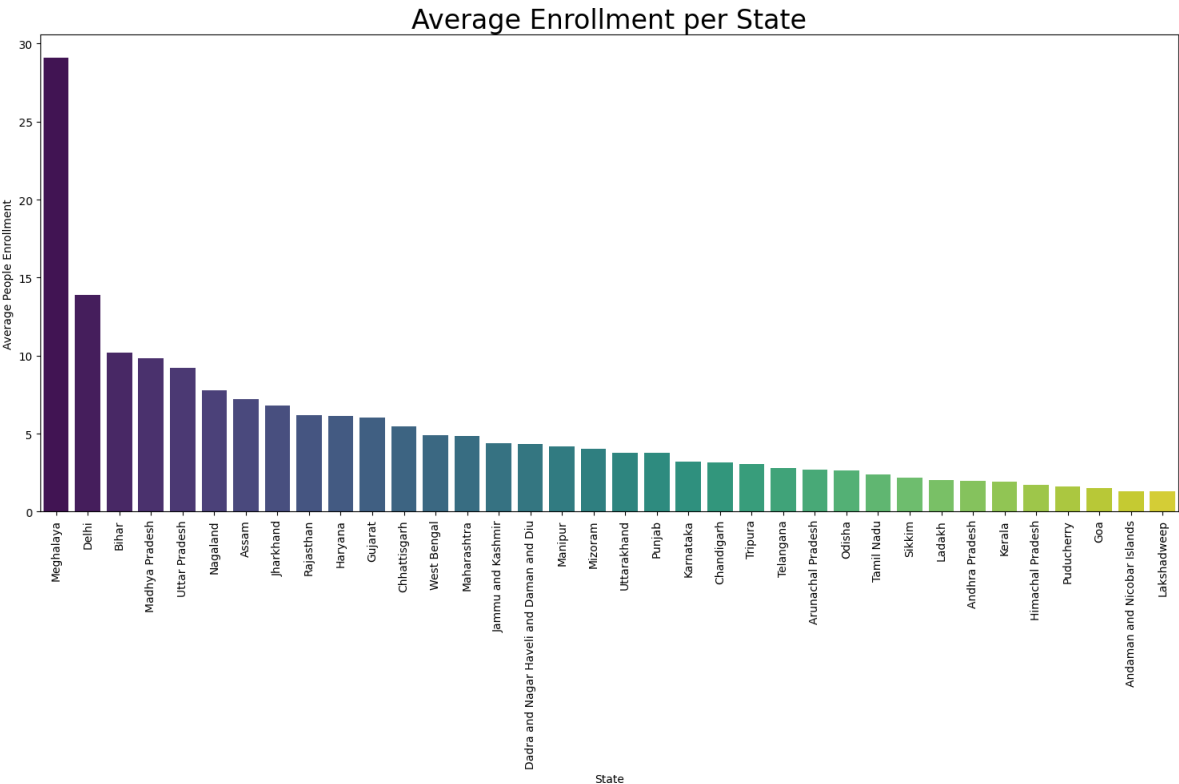


Figure 4 — Average daily enrollment per state

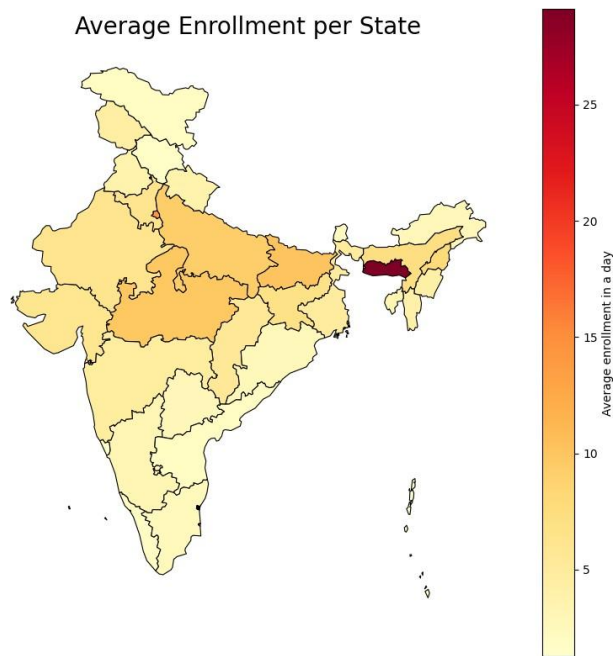


Figure 5 — Geographic distribution of average enrollment across India

Urban–Rural Disparity

District-level analysis reveals a pronounced urban-rural divide. Using Telangana as an example: Hyderabad records the highest average enrollment among its districts, driven by greater urban awareness and easier access to services. Rural districts, in contrast, show substantially lower participation — pointing to gaps in accessibility and outreach infrastructure.

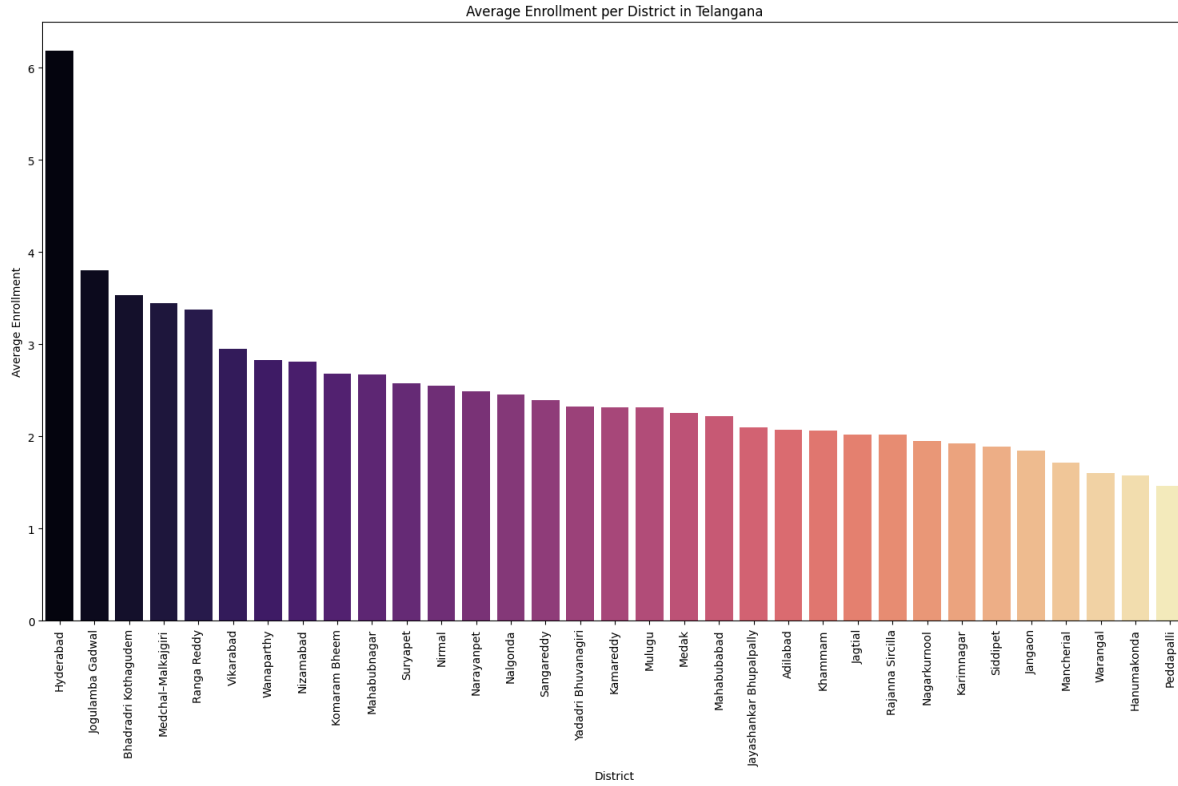


Figure 6 — Average enrollment by district in Telangana

Similar patterns are observed in Tamil Nadu, where Dharmapuri, Madurai, and Tiruvallur lead district-level performance, while many other districts lag, indicating that service availability is not uniform even within a single state.

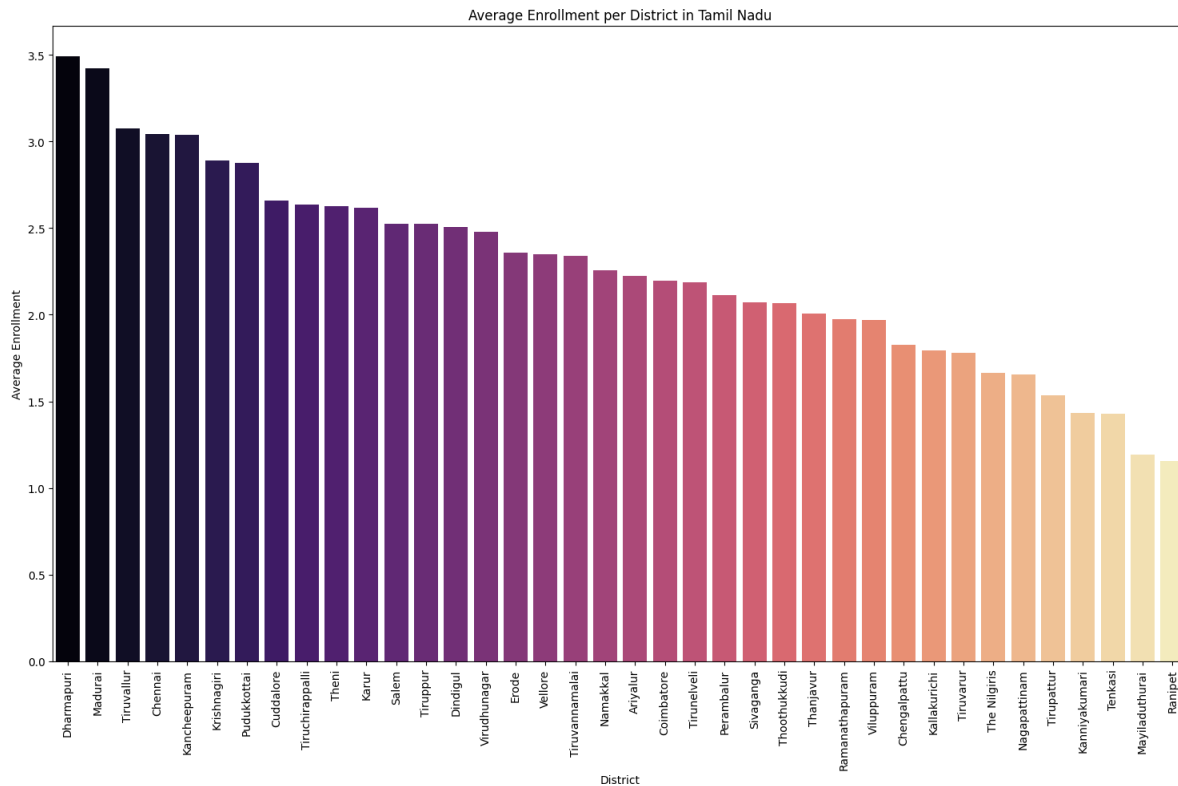


Figure 7 — Average enrollment by district in Tamil Nadu

Age-Group Enrollment Trends

Age Group	Observation	Implication
0-5 Years	Highest and most evenly spread enrollment across states	Early-age registration is relatively effective, possibly linked to birth-registration programmes
5-17 Years	Noticeable drop compared to 0-5 group	Many children miss follow-up enrollments, despite this being a critical age for education-related identity services
18+ Years	Very low in almost all states	Adult enrollment and biometric updates are largely neglected, indicating systemic under-participation among working-age adults

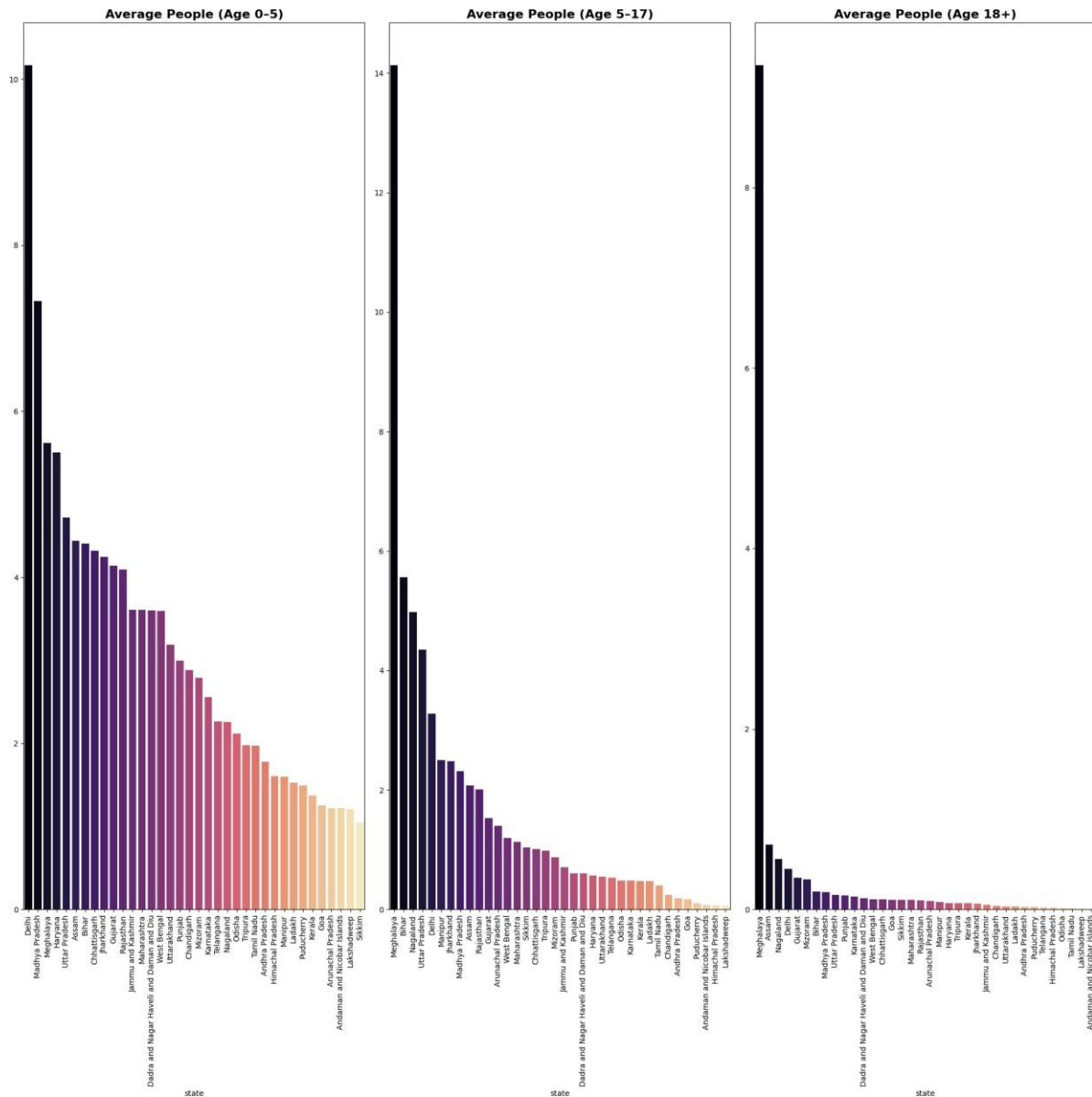


Figure 8 — Average enrollment by age group (0–5, 5–17, 18+) per state

4.2 Biometric Analysis

State-Level Biometric Activity

Delhi records significantly higher total biometric updates compared to all other states, followed by Madhya Pradesh and Chhattisgarh. This suggests better access to biometric facilities and higher population awareness in these regions. However, the majority of states show low biometric activity, risking authentication failures for residents with outdated or missing biometric data.

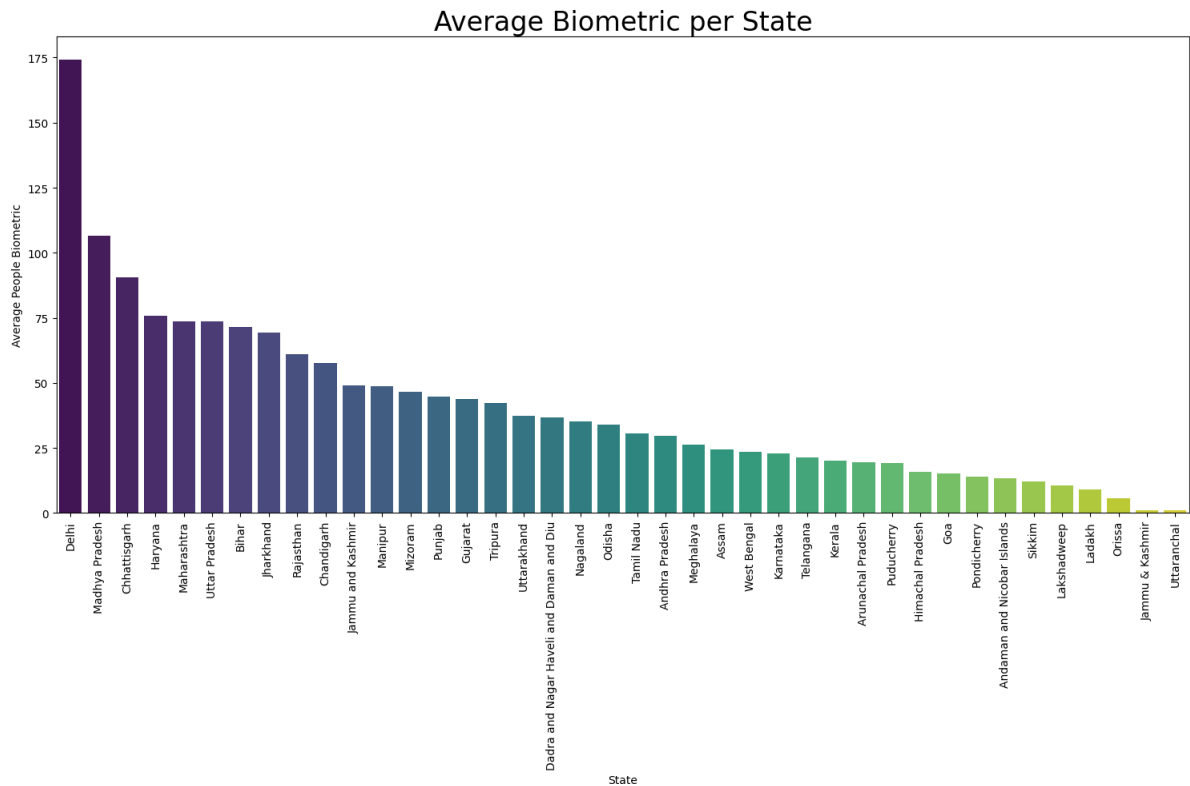


Figure 9 — Average biometric updates per state

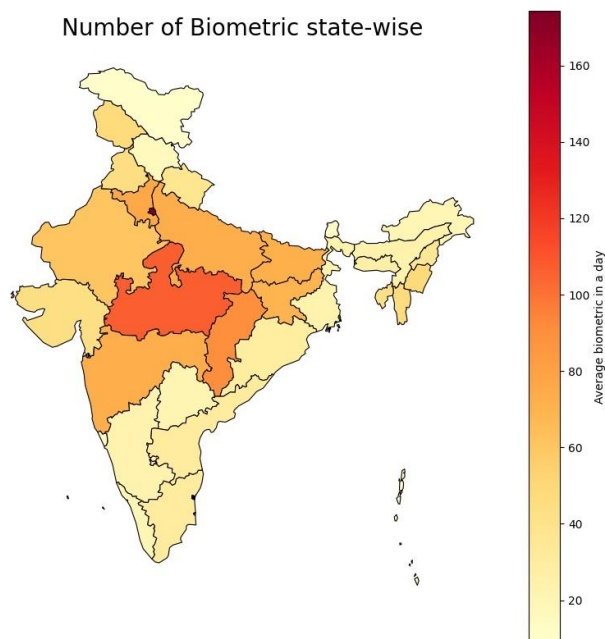


Figure 10 — Geographic distribution of biometric updates across India

District-Level Biometric Distribution (Delhi)

Within Delhi itself, biometric updates are unevenly distributed. Districts such as North West, South Delhi, and North East Delhi show very high biometric activity, whereas Shahdara, New Delhi, and

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4.3 Demographic Analysis

High-Demand States

States including Delhi, Uttar Pradesh, Bihar, Madhya Pradesh, Rajasthan, and Maharashtra exhibit very high Aadhaar activity volumes. This creates operational pressure — crowded enrollment centres, long queues, and extended waiting periods — that can actively discourage timely updates and renewals.

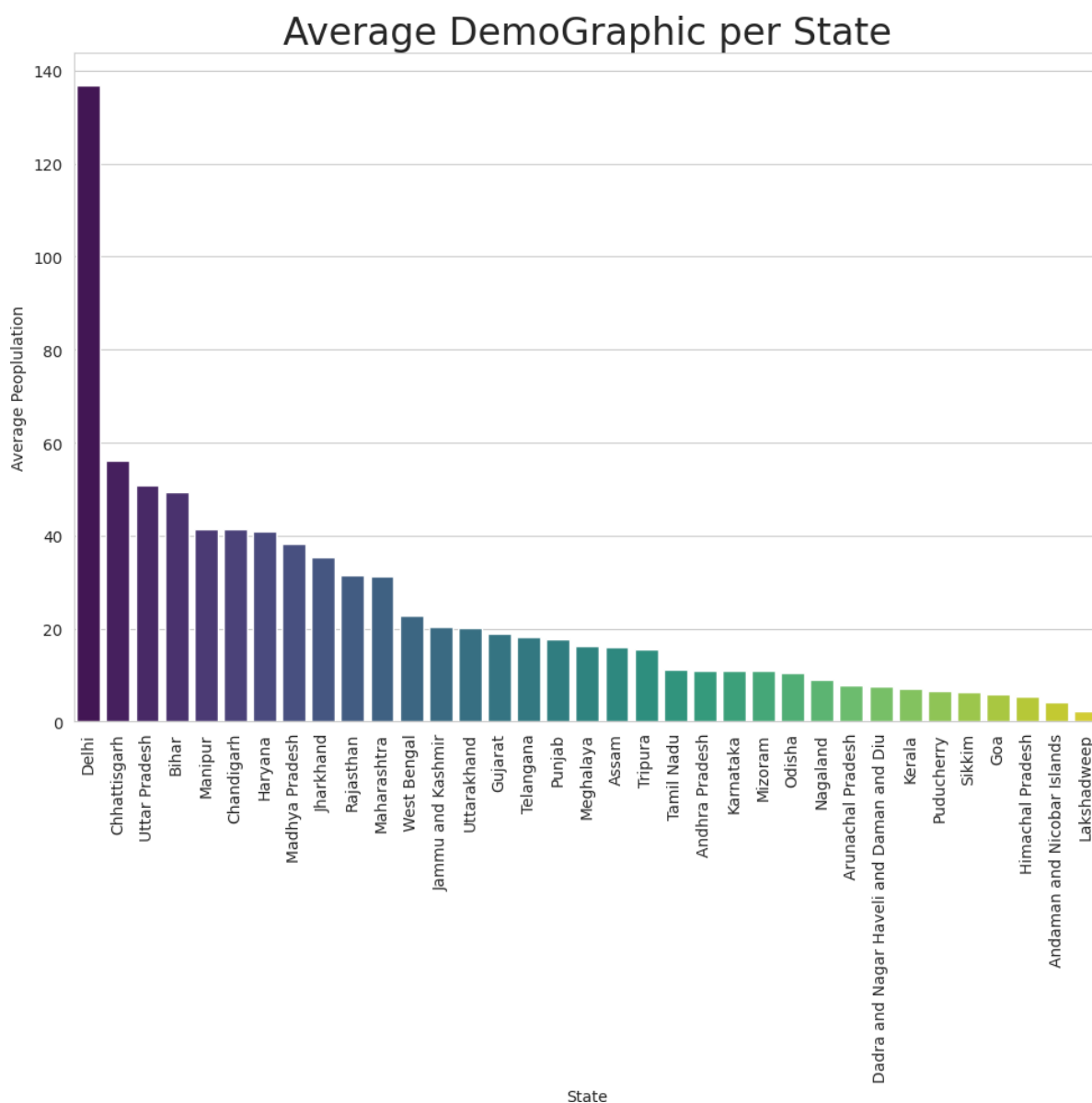


Figure 13 — Average demographic update activity per state

Low-Access States

Sikkim and north-eastern states generally show good population awareness, but their primary challenge is geographic: limited UIDAI centre availability and long travel distances. Critically, both extremes face exclusion risk — high-density states due to service overload and low-density states due to inaccessibility.

This analysis demonstrates that Aadhaar challenges are fundamentally operational in nature, not awareness-driven, and therefore require region-specific, tailored solutions rather than one-size-fits-all interventions.

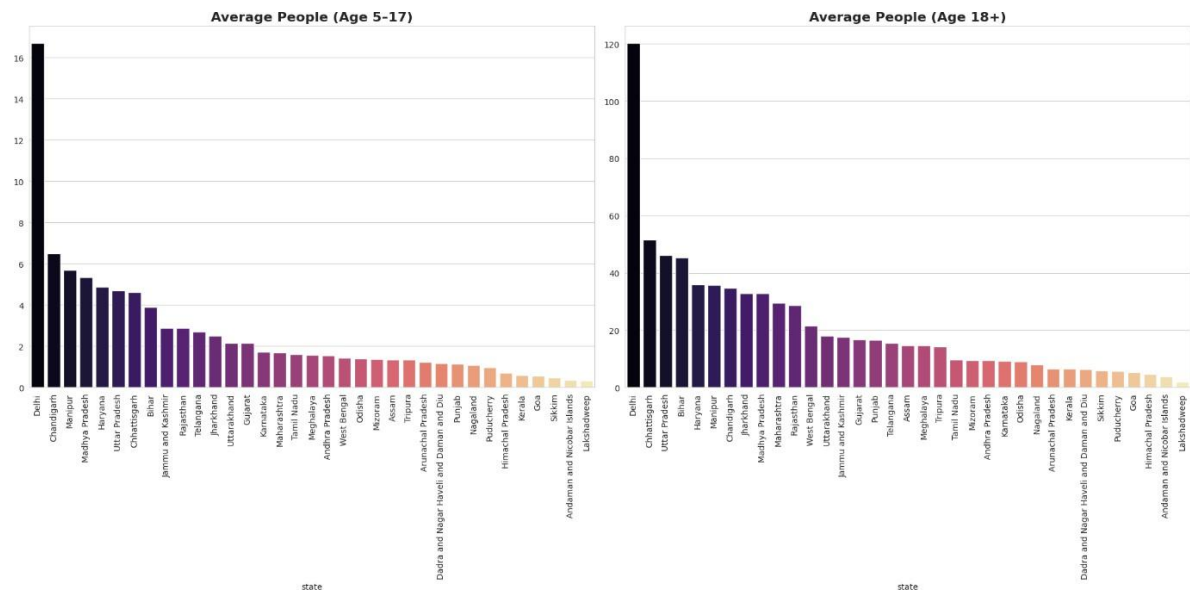


Figure 14 — Average demographic updates by age group (5–17 and 18+) per state

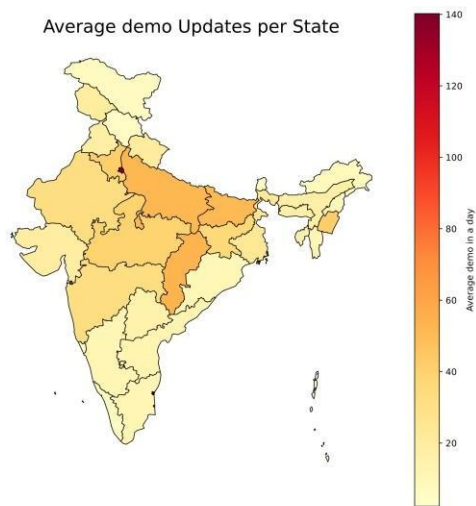


Figure 15 — Geographic distribution of demographic update activity across India

5. Policy Recommendations

5.1 Enrollment Improvements

1	Prioritise states and districts with low enrollment by deploying dedicated outreach teams and additional enrollment centres, rather than applying uniform national strategies.
2	Deploy mobile enrollment vans in rural, remote, and underserved areas to eliminate the travel barrier — a primary deterrent in low-density regions.
3	Develop age-group-specific campaigns: targeted school drives for children aged 5-17, and workplace or weekend camps for working adults to address the sharp decline in adult participation.
4	Study and replicate the successful practices of high-performing states like Meghalaya — analysing centre density, community partnerships, and outreach models.
5	Conduct awareness campaigns in regional languages using community leaders, local radio, and social media to communicate the practical benefits of Aadhaar enrollment.
6	Establish real-time monitoring dashboards to identify low-participation districts early and enable rapid, targeted responses.

5.2 Biometric Update Improvements

1	Launch regular biometric update drives, especially in states where biometric activity is critically low, coordinating with state administrations.
2	Introduce school-based biometric update camps for children aged 5-17, integrated with school enrollment or examination calendars.
3	Organise workplace and weekend biometric camps to reduce the friction of scheduling updates for the working adult population (18+).
4	Deploy mobile biometric units in rural and remote areas to ensure geographic coverage does not determine access.
5	Use SMS reminders, local language messaging, and community health workers to prompt residents about mandatory biometric update timelines.
6	Systematically adopt the best practices from high-performing states and scale them to low-performing regions through structured knowledge transfer programmes.

5.3 Demographic Update Improvements

1	Decongest high-demand centres in states like Delhi, Uttar Pradesh, Bihar, and Maharashtra by adding extra counters, extending operating hours, and introducing an online appointment/slot-booking system to reduce queue times.
2	Expand UIDAI centre density in low-access, high-awareness states such as Sikkim and the north-eastern states, where the bottleneck is geographic reach rather than demand.
3	Deploy mobile demographic-update units on a rotating circuit through remote districts so residents are not forced to travel long distances for routine name, address, or date-of-birth corrections.
4	Set staffing and resource allocation using actual demand data rather than uniform norms — high-volume states need more capacity, not just more centres.
5	Introduce a tiered token/priority system so that urgent corrections (e.g., address changes tied to welfare scheme eligibility) are not delayed behind routine updates.
6	Track average wait-time and update-turnaround as a public KPI per state to create accountability and guide future infrastructure investment.

6 Tools & Technologies Used

Category	Tools / Libraries
Programming Language	Python
Data Manipulation	NumPy, Pandas
Data Visualisation	Matplotlib, Seaborn, Plotly
Geospatial Analysis	GeoPandas
Data Source	UIDAI Official API (Enrollment, Demographic, Biometric datasets)

7 Conclusion

This analysis of UIDAI data reveals that Aadhaar enrollment and update services in India are characterised by significant and persistent regional, demographic, and operational inequalities. While certain states and urban districts demonstrate strong participation, a large portion of India's population — particularly in rural areas, north-eastern states, and specific age groups — remains underserved.

The central insight of this research is that Aadhaar exclusion is not primarily a product of population unawareness, but rather of operational and infrastructural deficits. Addressing these deficits requires granular, data-driven, region-specific policy interventions — not broad national campaigns.

The recommendations outlined in this report — targeted outreach, mobile service deployment, school and workplace integration, and continuous data monitoring — constitute a practical roadmap for UIDAI and policymakers to build a more equitable Aadhaar ecosystem.

— End of Report —
